



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ANNAI PETRO TRADERS IOC DEALE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

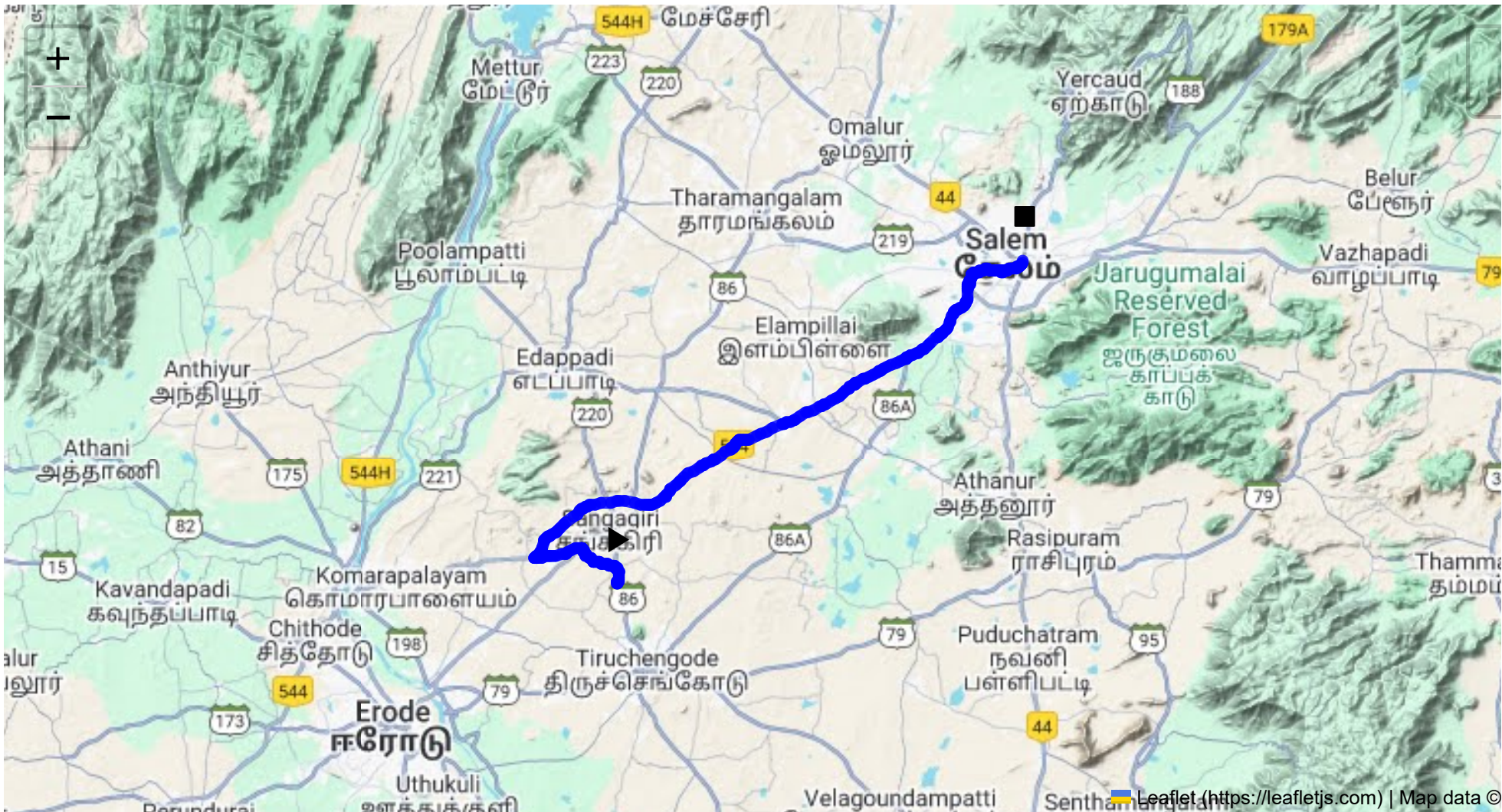
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 55.12 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.659301, 78.156937)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route connects Sangagiri in Tamil Nadu to Salem, navigating through several points such as Sankari RS, Bhavani Main Road, and Padaiveedu. It spans approximately 55 kilometers and follows mostly national highways and state roads, which are significant for connecting different industrial and rural areas.

2. Typical Weather Conditions and Potential Hazards

Tamil Nadu experiences tropical weather conditions throughout the year. Key considerations include:

- **Monsoon Season (June to September):** Heavy rains can cause flooding, slippery roads, and reduced visibility.
- **Summer (March to May):** Extreme heat can affect vehicle performance, particularly tires and engines.
- **Potential Hazards:** Flash floods during monsoons and heat-related damage during summer months.

3. Traffic Patterns

- **Peak Hours:** Typically from 8 AM to 10 AM and 5 PM to 7 PM. Congestion is common near towns and markets.
- **Congestion-Prone Areas:** Major junctions around Sankari and Salem, particularly near intersections with Bhavani Main Road.

4. Road Quality and Infrastructure

- **Road Quality:** Generally good on major roads like National Highways, but some rural roads may have potholes and uneven surfaces.
- **Infrastructure:** Adequate signage and lighting in urban areas; rural stretches might be less maintained.

5. Alternative Routes for Emergencies

- **Via NH544:** Another route through National Highway 544 can be taken in emergencies, offering a more direct and possibly faster path but with higher traffic.
- **Local Bypasses:** Utilizing local bypasses can help avoid congested areas during emergencies.

6. Local Regulations Affecting Hazardous Material Transport

- Restrictions on transportation times and specific permits are required for the movement of hazardous materials.
- Hazardous materials are usually not allowed during night time in certain segments of the route.

7. Historical Incidents

- Historically, there have been minor accidents involving heavy vehicles due to road conditions and traffic congestion, particularly near Salem.
- No significant hazardous material incidents have been recorded recently.

8. Environmental Considerations and Sensitive Areas

- **Ecological Zones:** Avoid idling near protected forest areas and agricultural zones to minimize pollution.

- **Water Bodies:** Routes running alongside rivers must be carefully navigated during monsoons.

9. Communication Coverage

- **Urban Areas:** Generally good coverage with 4G services available.
- **Rural Areas:** Potential dead zones in some remote segments; drivers should ensure their communication devices are well-charged and alternative communication methods are available.

10. Estimated Emergency Response Times

- **Urban Segments:** 15-30 minutes response time due to proximity to services.
- **Rural Segments:** Could extend to 45 minutes or more, depending on accessibility and weather conditions.

12. Overall Summary of Risk Assessment

This route is moderately safe for transporting hazardous materials with appropriate precautions. Weather conditions can present significant challenges especially during the monsoon and summer months. Urban congestion and potential communication dead zones require strategic planning. Continuous monitoring of local infrastructure improvements and maintaining up-to-date communication devices are vital for safety. Regular training for drivers in handling unexpected situations and familiarity with alternative routes could significantly mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.65514, 78.15419	15 KM/Hr
1	Roundabout	High	11.65653, 78.15533	15 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46048, 77.85036	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46318, 77.84913	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45634, 77.81673	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.61736, 78.10588	30 KM/Hr
13	Turn	Medium	11.61753, 78.10608	30 KM/Hr
14	Turn	Medium	11.64007, 78.11794	30 KM/Hr
15	Turn	Medium	11.64020, 78.11786	30 KM/Hr
16	Turn	Medium	11.64497, 78.12028	30 KM/Hr
17	Turn	High	11.64503, 78.12034	15 KM/Hr
18	Turn	High	11.64970, 78.12064	15 KM/Hr
19	Turn	Medium	11.64986, 78.12096	30 KM/Hr
20	Turn	Medium	11.65004, 78.12212	30 KM/Hr
21	Turn	Medium	11.65099, 78.12276	30 KM/Hr
22	Turn	Medium	11.65113, 78.12473	30 KM/Hr
23	Turn	Medium	11.65203, 78.14157	30 KM/Hr
24	Blind Spot	Blind Spot	11.65141, 78.14234	10 KM/Hr
25	Turn	High	11.65197, 78.14241	15 KM/Hr
26	Turn	Medium	11.65756, 78.15669	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Ayyappa Hospital	11.6524111, 78.1456836	30 km/h	Medium
16	hospital	Pramilagurunathan Hospital	11.6544063, 78.1464484	30 km/h	Medium
18	police	Shevapet B 2 Police Station	11.6514754, 78.1522608	30 km/h	Medium
19	hospital	Salem Government Hospital	11.6554386, 78.1520712	30 km/h	Medium
20	hospital	Corporation Maternity Hospital	11.6524437, 78.154675	30 km/h	Medium
24	hospital	Shri Suraksha Speciality Medical Centre	11.6590215, 78.1587125	30 km/h	Medium
25	hospital	Sri Sellappa Hospital	11.65978, 78.154787	30 km/h	Medium
26	hospital	A.G.P Hospital	11.6608027, 78.1554709	30 km/h	Medium
27	hospital	Kerrison Memorial Hospital	11.6604257, 78.1579573	30 km/h	Medium
28	clinic	Salem Heart Clinic	11.660386, 78.1556973	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	school	CSI Hr. Sec. School	11.6536296, 78.1529373	30 km/h	Medium
21	school	Govt Boys Hr. Sec. School	11.6550838, 78.155882	30 km/h	Medium
22	school	Govt Girls Hr Sec School	11.6550993, 78.1575234	30 km/h	Medium
23	school	Govt Girls Hr Sec School, Fort	11.6547568, 78.1578092	30 km/h	Medium
29	school	Corporation High School	11.6608771, 78.1573617	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65514, 78.15419



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65653, 78.15533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64007, 78.11794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64020, 78.11786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64497, 78.12028



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64503, 78.12034



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64970, 78.12064



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64986, 78.12096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65004, 78.12212



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65099, 78.12276



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65113, 78.12473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65203, 78.14157



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.65141, 78.14234



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65197, 78.14241



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65756, 78.15669